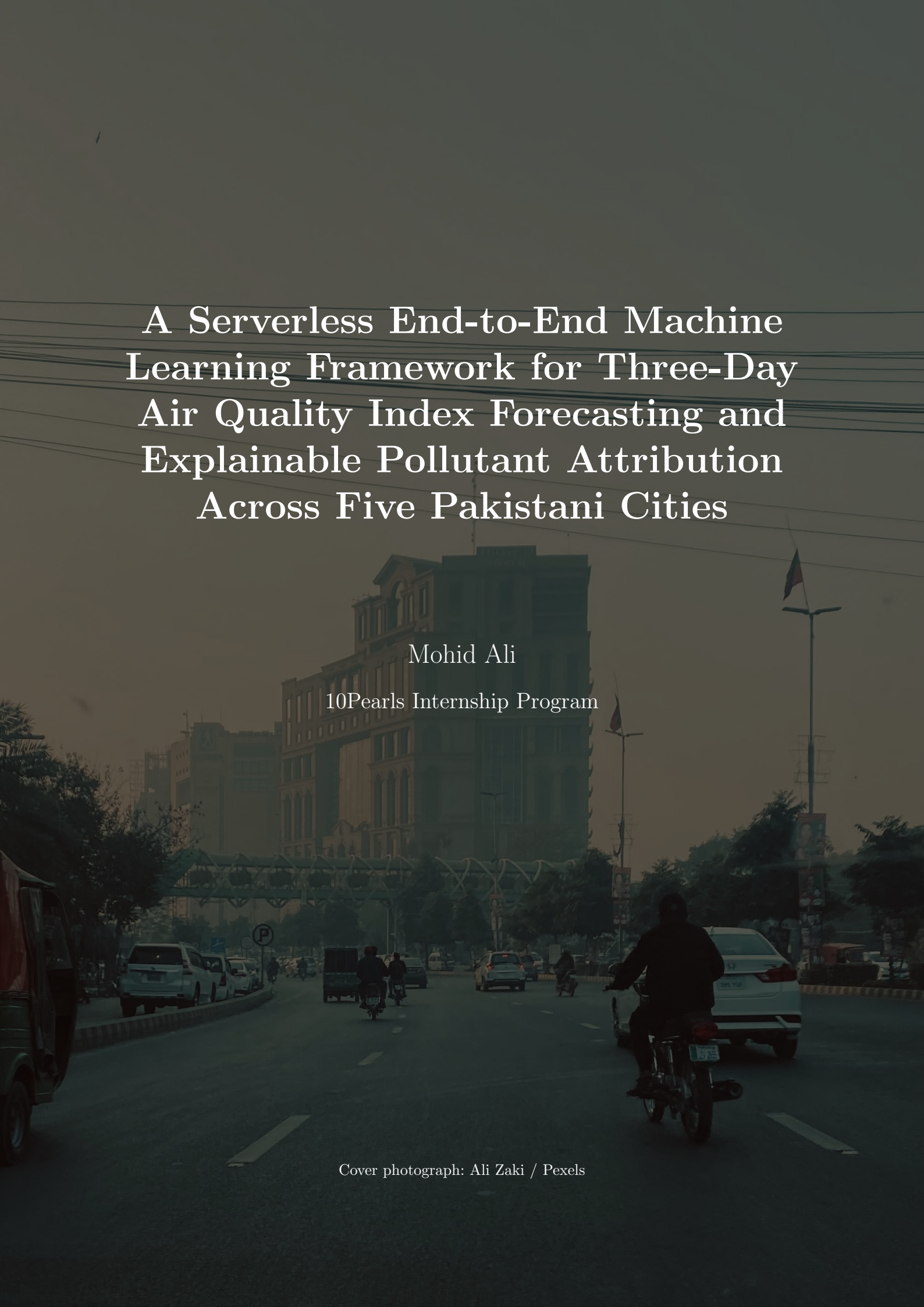




A Serverless End-to-End Machine Learning Framework for Three-Day Air Quality Index Forecasting and Explainable Pollutant Attribution Across Five Pakistani Cities

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Abstract

Air pollution is a persistent public health hazard in Pakistan, yet residents have no accessible way to know what the air will look like beyond a same-day reading. This project builds a fully serverless, end-to-end machine learning system that forecasts the United States Environmental Protection Agency (EPA) Air Quality Index three days ahead for five major Pakistani cities – Karachi, Lahore, Faisalabad, Islamabad, and Peshawar. Rather than relying on OpenWeather’s coarse five-level air quality index, the EPA AQI is computed directly from raw pollutant concentrations using unit conversion and the EPA’s own piecewise-linear breakpoint table, giving a continuous 0–500 score comparable to public references such as aqicn.org. Three years of hourly pollution data from the OpenWeather Air Pollution API and weather data from Open-Meteo are cleaned and engineered into hourly and daily feature tables, both served through a Hopsworks feature store that is refreshed automatically every hour by a GitHub Actions pipeline. Six forecasting approaches – persistence, ridge regression, random forest, XGBoost, per-city SARIMAX, and a two-layer LSTM – are trained on a strictly chronological split and compared against a persistence baseline at one, two, and three days ahead. The random forest is the strongest model at every horizon and is registered in a Hopsworks model registry, with a daily retraining pipeline that only replaces the production model if a candidate genuinely improves on it. SHAP analysis of the winning model finds that same-day mean PM2.5 dominates the three-day forecast by more than five times the next most important feature, consistent with PM2.5 being the dominant pollutant in the large majority of observed hours. The complete system, including live forecasts, a colour-coded hazard alert, a thirty-day trend chart, and a model explainability view, is exposed through a public Streamlit dashboard.

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1 Introduction

Air pollution is one of Pakistan’s most severe and least forecast public health hazards. Lahore in particular is regularly ranked among the most polluted cities in the world during its October-to-February smog season, when crop burning, vehicle emissions, and winter temperature inversions combine to trap fine particulate matter close to the ground. Existing public tools such as aqicn.org and OpenWeather’s own air pollution API report only the current or, at best, same-day pollutant readings, and OpenWeather additionally compresses its own air quality signal into a coarse five-level integer index that is too imprecise to support quantitative forecasting. Residents, schools, and outdoor workers in these cities have no accessible way to know whether the air three days from now will be merely unhealthy or genuinely hazardous.

This project addresses that gap with a system built around three questions: what is the air quality right now, what will it be over the next three days, and what is driving that forecast. Answering them required building, rather than borrowing, a proper air quality index: the EPA’s 0–500 AQI is computed directly from raw pollutant concentrations following the EPA’s own published methodology, rather than taken from any third-party source. The resulting hourly and daily feature tables are used to train and compare six forecasting approaches, the best of which is explained with SHAP and served through a live, publicly accessible dashboard, with the entire system, from an hourly data refresh to a daily model retrain, running without a server the author has to manage.

2 Study Area and Data

2.1 Study Area

Five of Pakistan’s largest cities were selected to give geographic and climatic spread: the coastal megacity of Karachi in the south, the agricultural Punjab plain cities of Lahore and Faisalabad, and the elevated cities of Islamabad and Peshawar near the northern hills, the latter two also close to regions affected by seasonal crop burning.

City	Latitude	Longitude
Karachi	24.8607	67.0011
Lahore	31.5204	74.3587
Faisalabad	31.4504	73.1350
Islamabad	33.6844	73.0479
Peshawar	34.0151	71.5249

Table 1: Study cities and their coordinates. All timestamps are stored in UTC and converted to Pakistan Standard Time, UTC+5 with no daylight saving, only for display.

2.2 Data Sources

Data was assembled from two public sources, chosen so that pollution and weather could each come from the provider best suited to it:

- **OpenWeather Air Pollution API:** current, history (available from 27 November 2020), and forecast endpoints, returning eight pollutant components – CO, NO,

NO₂, O₃, SO₂, PM2.5, PM10, and NH₃ – in µg/m³, along with OpenWeather’s own coarse 1–5 air quality index, which is not used for modelling.

- **Open-Meteo**: the ERA5 reanalysis archive endpoint for historical backfill, and the forecast endpoint, which supplies up to seven days of genuine forecast weather with no API key required. Hourly variables used are temperature, relative humidity, dew point, precipitation, surface pressure, wind speed, wind direction, wind gusts, and cloud cover.

2.3 Master Dataset Schema

All sources are merged into two feature groups in the Hopsworks feature store, at an hourly and a daily grain:

- **aqi_hourly** (primary key city, timestamp_utc): 33 columns spanning the eight raw pollutant concentrations, nine weather variables, the computed `epa_aqi` and `dominant_pollutant`, and cyclical sine/cosine encodings of hour, day of week, month, and wind direction.
- **aqi_daily** (primary key city, date): 63 columns aggregating the hourly table to a daily mean and max per pollutant and a daily weather mean, four AQI lag features and three PM2.5 lag features, rolling AQI mean and standard deviation, a PM2.5-to-PM10 ratio, one-hot city indicators, cyclical calendar features, the three forecast targets `aqi_d1`, `aqi_d2`, and `aqi_d3`, and the matching next-1/2/3-day forecast weather features `fc_temp`, `fc_wind`, `fc_precip`, and `fc_humidity`.

A three-year backfill (28 July 2023 to 27 July 2026) across all five cities produced 127,992 raw hourly pollution rows and 131,520 raw hourly weather rows, with 2.5 to 2.65 percent of pollution hours missing per city spread across 21 to 22 small gaps, and zero missing weather values.

2.4 Data Preprocessing

Pollution and weather were outer-merged on city and timestamp so that a gap in one source does not drop rows the other still has, then each city was reindexed onto a complete hourly range and interpolated with a limit of three hours, leaving gaps longer than that as missing rather than guessed at. Outliers were reviewed but deliberately not clipped: per-city PM2.5 maxima ranged from 517 to 1308 µg/m³, and these were confirmed to already be present in the raw pulled data before any merging or reindexing, making them genuine CAMS atmospheric model extremes rather than an artefact of processing.

3 Methodology

3.1 EPA AQI Calculation

This is the central technical contribution of the project. OpenWeather’s own air quality field is a coarse integer from 1 to 5, far too imprecise to regress on or to compare against public references. Instead, the US EPA AQI (0–500) is computed directly from the raw pollutant concentrations, following the EPA’s Technical Assistance Document for the

Reporting of Daily Air Quality, EPA-403/B-26-003 [3]: the four gaseous pollutants are converted from $\mu\text{g}/\text{m}^3$ to ppb or ppm at 25°C, each pollutant is averaged over the window the EPA specifies – 24 hours for PM2.5 and PM10, 8 hours for O₃ and CO, and 1 hour for SO₂ and NO₂ – and each averaged concentration is truncated to the EPA’s specified number of decimal places before a piecewise-linear interpolation is applied to the official breakpoint table. The overall AQI is the maximum of the six pollutant sub-indices, and the pollutant that produced it is recorded as the dominant pollutant, a field almost always occupied by PM2.5 in this dataset.

3.2 Feature Engineering

The hourly feature table adds the computed `epa_aqi` and `dominant_pollutant` alongside cyclical time encodings to the cleaned hourly data. This is then aggregated to a daily table carrying lag features, rolling statistics, and the three forecast targets. The daily table also carries the next three days of weather as known-future features, since wind, temperature, and precipitation forecasts are legitimately available at prediction time and are strong predictors of how pollution disperses. During backfill this column is filled from the ERA5 archive as a stand-in for a genuine forecast, which is disclosed as a limitation in Section 8; the live-serving dashboard instead queries Open-Meteo’s real forecast endpoint at request time.

3.3 Predictive Modelling

Six approaches were trained and compared, one global model per approach across all five cities rather than five separate per-city models, with city included as a one-hot feature. This gives five times the training rows, lets the model learn shared weather-to-AQI physics, and produces a single artifact to register and serve, while per-city metrics are still reported so a model cannot hide a badly performing city.

- **Persistence:** tomorrow’s AQI equals today’s, the baseline every other model must beat.
- **Ridge regression:** a linear baseline over the full feature set.
- **Random forest:** hyperparameters tuned with Optuna over `TimeSeriesSplit` folds, minimising mean absolute error natively across its three horizon outputs.
- **XGBoost:** also Optuna-tuned, fitting one booster per horizon since XGBoost has no native multi-output mode.
- **SARIMAX:** fit per city on the univariate AQI series alone.
- **LSTM:** a two-layer recurrent network consuming the past 72 hours of 15 hourly features per city to forecast all three horizons jointly.

3.4 Model Explainability

The winning model, random forest, is explained with a SHAP `TreeExplainer`, focused on the one-day-ahead forecast since that is the horizon closest to today and the one the dashboard shows first.

3.5 Automation and Deployment

Two GitHub Actions workflows keep the system live: an hourly pipeline that fetches a recent window of pollution and weather, recomputes the EPA AQI and engineered features, and upserts both feature groups, and a daily pipeline that retrains the random forest on the current feature store contents and only promotes a new model version to the Hopsworks model registry if it beats the currently registered one on one-day-ahead RMSE. Both pipelines are idempotent: Hudi upserts by primary key mean recomputing the same city and hour twice overwrites with the same values rather than duplicating rows.

4 Evaluation Metrics

All models use a strictly chronological split: the test set is the most recent 60 days (26 May 2026 to 28 July 2026), and hyperparameter tuning validates on `TimeSeriesSplit` folds within the training period, sorted by date across all five cities so a fold is a genuine calendar cutoff rather than a fold across cities. A random train/test split was deliberately avoided, since shuffling lagged time-series data lets the model see rows adjacent to its own test rows during training and produces misleadingly optimistic scores. After dropping rows with a missing feature or target, mostly the last three days of each city’s series where the three-day-ahead target cannot yet exist, 4,190 rows remain for training and 224 for testing. Models are scored with root mean squared error (RMSE), mean absolute error (MAE), and R^2 , computed per city and per horizon and then averaged across the five cities for comparison.

5 Results Summary

5.1 Predictive Model Performance

Table 2 reports RMSE for each model at each forecast horizon, averaged across the five cities. Random forest, XGBoost, SARIMAX, and the LSTM all beat the persistence baseline at every horizon. Ridge does not: it edges persistence at one day ahead (17.69 against 18.45) but loses to it at two and three days ahead, since a linear model fit on 58 mostly collinear pollutant, weather, and lag columns has far less capacity than a tree ensemble or recurrent network to correct itself multiple days out. Random forest is the strongest model at every horizon and is the model carried forward to explainability and the production model registry.

Model	aqi_d1	aqi_d2	aqi_d3
Persistence	18.45	26.63	29.56
Ridge	17.69	27.70	31.01
Random Forest	10.50	20.45	23.52
XGBoost	12.79	23.46	23.99
SARIMAX	17.51	24.27	26.08
LSTM	13.10	22.82	26.65

Table 2: RMSE by model and forecast horizon, averaged across all five cities on the 60-day chronological test set.

5.2 Exploratory Findings

PM2.5 is the dominant pollutant in 119,714 of 125,752 hourly readings with a computable AQI, a little over 95 percent of the time. Lahore’s mean AQI more than doubles in the October-to-February smog season compared to the rest of the year, rising from 144.6 to 339.5, and Lahore recorded zero hours of Good air quality across the full three-year backfill.

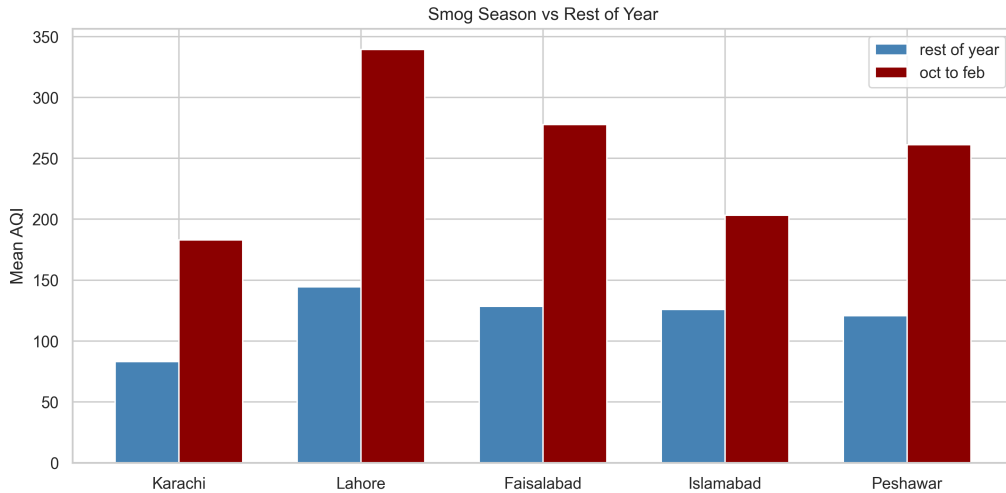


Figure 1: Mean AQI in the October-to-February smog season against the rest of the year, by city.

6 Model Interpretability

SHAP values for the random forest’s one-day-ahead forecast show that today’s mean PM2.5 dominates the prediction outright, contributing more than five times the next most important feature, today’s maximum PM2.5, and far ahead of the AQI lag and rolling features or the forecast wind and humidity columns that were expected to matter more. This is consistent with the exploratory finding that PM2.5 is already the dominant pollutant in the large majority of hours, so today’s raw reading carries more signal than anything already derived from it.

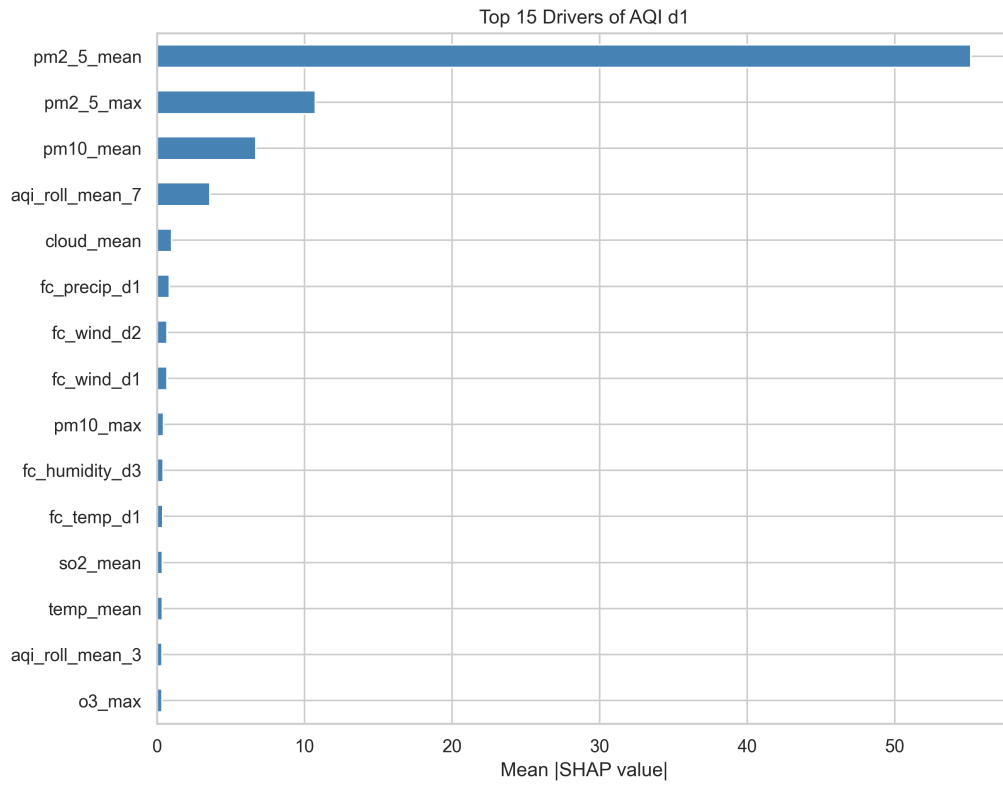


Figure 2: Mean absolute SHAP value per feature for the random forest’s one-day-ahead forecast, ranking the strongest predictors of tomorrow’s AQI.

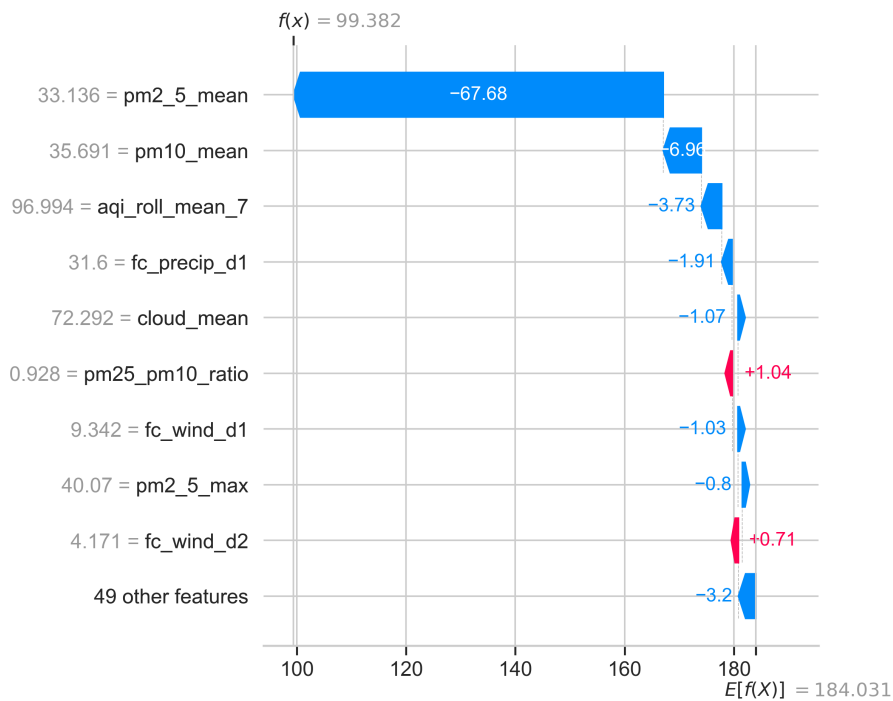


Figure 3: SHAP waterfall breakdown for a single Lahore prediction, showing how each feature moves the forecast away from the model’s average output.

7 Streamlit Dashboard

The full system – current conditions, the three-day forecast, and the model’s explainability and performance – is exposed through a live dashboard built with Streamlit and deployed publicly at <https://pearls-aqi-predictor-mohid.streamlit.app>. A sidebar lets the user choose any of the five cities. The main view shows a gauge for the current AQI shaded with the six EPA category bands, the category name and dominant pollutant, live readings for PM2.5, PM10, O₃, and NO₂, and a colour-coded hazard alert banner – green, orange, red, or maroon – carrying EPA health guidance text matched to the current category. Below this, three forecast cards show the predicted AQI and category for each of the next three days, followed by a 30-day trend chart with the same six EPA bands shaded behind both the actual and forecast series, and a comparison section showing all five cities’ current AQI on a map and a ranked bar chart. Two tabs complete the page: an Explainability tab showing a live SHAP waterfall for the selected city’s next-day prediction, and a Model performance tab showing the RMSE comparison table from Section 2 broken down per city.

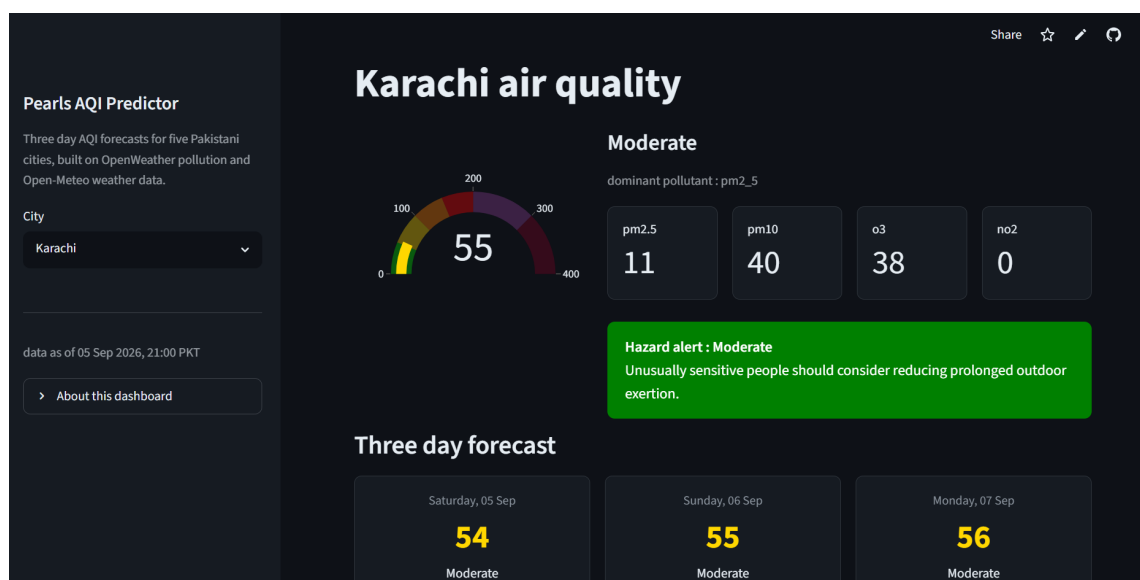


Figure 4: Dashboard overview for Karachi, showing the AQI gauge, hazard alert banner, and three-day forecast cards.

8 Ethics and Limitations

This project is an educational and portfolio system, not a regulatory air quality monitor, and its forecasts should not be the sole basis for a health decision in genuinely hazardous conditions; a real emergency warrants checking an official government or embassy air quality source directly.

First, OpenWeather’s pollution data comes from the CAMS atmospheric model rather than ground stations. It reads noticeably smoother than a physical monitor and does not match aqicn.org exactly, a known and disclosed limitation rather than a bug in this system.

Second, the known-future weather features used during backfill and training come from the ERA5 archive rather than a genuine forecast, since the archive is only available after

the fact. This makes backtest scores slightly optimistic relative to true live performance, since the model effectively saw perfect hindsight weather during training. The deployed dashboard closes this gap for real predictions by querying Open-Meteo’s actual forecast endpoint at request time, but the reported model comparison metrics in Table 2 are still backtest scores subject to this caveat.

Third, GitHub Actions scheduled workflows are best-effort and can run five to twenty minutes late, and are automatically disabled by GitHub after 60 days of repository inactivity, both of which are operational risks for a system meant to update hourly.

Fourth, Hopsworks’ free tier constrains the deployment: only one application may be deployed from a private GitHub repository on Streamlit Community Cloud’s free tier, which is why this project’s repository is public rather than private, a deliberate tradeoff of source code visibility for zero-cost hosting rather than an oversight.

Finally, ridge regression underperforming the persistence baseline at two and three days ahead is reported plainly in Table 2 rather than omitted, since it is a genuine and informative finding about the limits of a linear model on this feature set.

9 Conclusion

This project delivers a fully serverless, end-to-end air quality forecasting system for five major Pakistani cities, combining a properly computed EPA air quality index, automated hourly and daily pipelines, a random forest that outperforms five alternative models including a persistence baseline and an LSTM, and an explainability layer that identifies same-day PM_{2.5} as the dominant driver of the three-day forecast. The system is exposed through a public, interactive dashboard rather than left as a notebook exercise, closing the loop between analysis and something a resident of one of these cities could actually check before stepping outside. Future work includes a FastAPI `/predict` endpoint to complement the dashboard, confirming that the hourly and daily GitHub Actions workflows continue to fire on their own schedule rather than only on manual triggers, and extending the underlying feature store to additional cities.

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